PYGEOVISION · GEOSPATIAL ANALYSIS REPORT

# Forest Change (JAXA PALSAR forest / non-forest)

2018 to 2020 · 3,039.1 km²

| 171,519.6 ha | 180,995.9 ha | 6,820.8 ha | 16,297.0 ha |
|---|---|---|---|
| Forest, first year | Forest, second year | Forest loss | Forest gain |

## Executive summary

- Forest went from 171,519.6 ha in 2018 to 180,995.9 ha in 2020: loss 6,820.8 ha, gain 16,297.0 ha.
- The loss is 4.0 % of the 2018 forest (3,410.4 ha per year).

JAXA global PALSAR/PALSAR-2 forest/non-forest map (25 m, L-band radar, yearly 2015-2020; in this product 'forest' is natural forest with at least 90% canopy cover unless include_open_forest is set). Each year is mapped separately, so gains and losses include mapping disagreement; plantations count as forest.

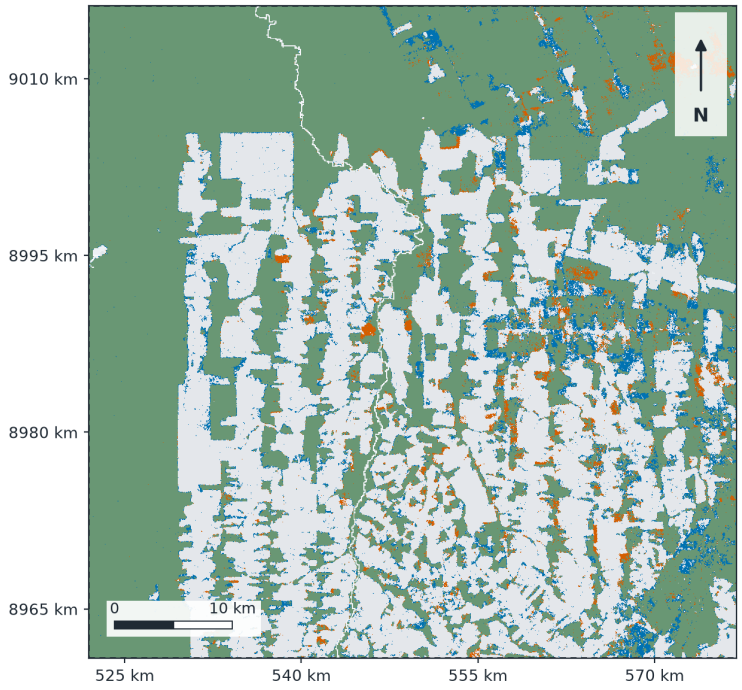
## Study area and inputs

| | |
|---|---|
| Area of interest (west, south, east, north) | -62.8000, -9.4000, -62.3000, -8.9000 |
| Centre | -9.1500°, -62.5500° |
| Area of interest | 3,039.15 km² |
| Period | 2018 to 2020 |
| Output grid | 2,201 × 2,215 pixels, EPSG:32720 |
| Raster extent area | 3,047.01 km² |
| Valid (cloud-free) share | 99.6 % |

# Results

## Forest Change (JAXA PALSAR forest / non-forest)

Forest change · 2018 to 2020 · AOI -62.8000, -9.4000, -62.3000, -8.9000




**Legend**

- **Non-forest** 115,784.6 ha · 38.1 %
- **Forest in both years** 164,698.9 ha · 54.2 %
- **Forest loss** 6,820.8 ha · 2.2 %
- **Forest gain** 16,297.0 ha · 5.4 %

**Key figures**

Forest, first year
**171,519.6 ha**

Forest, second year
**180,995.9 ha**

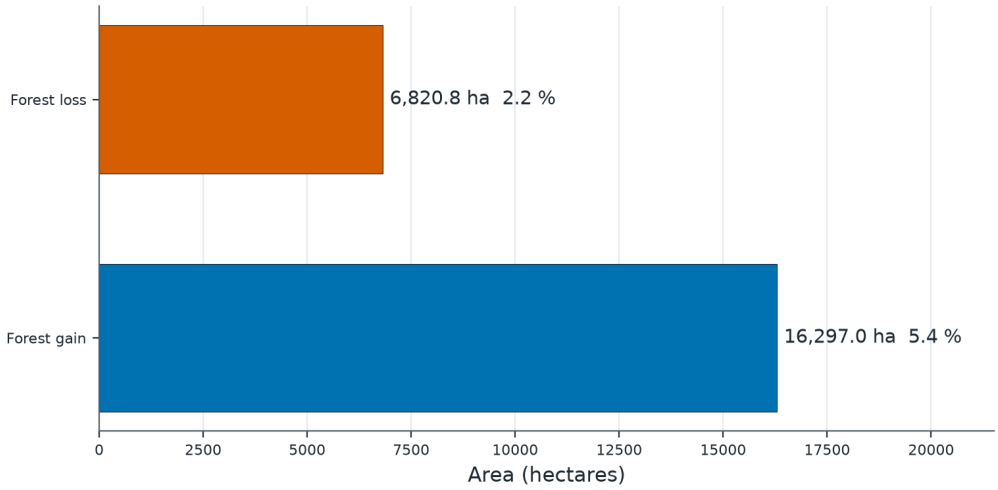
Forest loss
**6,820.8 ha**

Forest gain
**16,297.0 ha**

CRS: EPSG:32720 · Areas measured from the output raster · No basemap · PyGeoVision 2.2.0 · 2026-10-17 19:27 UTC

*Map: Forest Change (JAXA PALSAR forest / non-forest)*

## Area by class: Forest change




*Summary chart*

### Class areas: Forest change

| Class | Value | Pixels | Area (ha) | Share of valid area | Share of AOI |
|---|---|---|---|---|---|
| Non-forest | 0 | 1,852,554 | 115,784.6 | 38.1 % | 38.0 % |
| Forest in both years | 1 | 2,635,182 | 164,698.9 | 54.2 % | 54.1 % |
| Forest loss | 2 | 109,132 | 6,820.8 | 2.2 % | 2.2 % |
| Forest gain | 3 | 260,752 | 16,297.0 | 5.4 % | 5.3 % |

*Valid area 303,601.2 ha of 304,700.9 ha (4,857,620 of 4,875,215 pixels). Areas are measured from the output raster.*

### Key statistics (reported by the pipeline)

| Measure | Value |
|---|---|
| Forest, first year | 171,519.6 ha |
| Forest, second year | 180,995.9 ha |
| Forest loss | 6,820.8 ha |
| Forest gain | 16,297.0 ha |
| Net change | 9,476.2 ha |
| Loss per year | 3,410.4 ha |
| Loss as share of first-year forest | 4.0 % |
| Valid share of the area | 99.6 % |

### Other statistics

| Measure | Value |
|---|---|
| Year before | 2,018 |
| Year after | 2,020 |
| Forest share before | 56.5 % |
| Forest share after | 59.6 % |
| Include open forest | no |
| Resolution m | 25.000 |
| Resolution coarsened | no |

# Method and parameters

How much forest was lost or gained between two years from the radar-based forest map?

The JAXA global PALSAR forest / non-forest map (25 m, 2015-2020) is read for both years from the Microsoft Planetary Computer and resampled onto one UTM grid.

'Forest' is class 1 (canopy cover above 90 %) by default, or classes 1 and 2 (above 10 %) with include_open_forest.

Pixels that are water or no-data in either year are nodata. Loss is forest then, non-forest now; gain is the reverse.

| | |
|---|---|
| Pipeline | forest_change |
| Area of interest | -62.8000, -9.4000, -62.3000, -8.9000 |
| cloud_mask | True |
| date_after | 2020 |
| date_before | 2018 |
| period | 2018 to 2020 |

# Interpretation and limitations

How to read the result

- Radar backscatter responds to structure and moisture, so plantations count as forest and flooded or very dry conditions can move pixels between classes. Treat small, scattered changes as noise.

Limitations

- Each year is mapped separately: gains and losses include mapping disagreement, not only change on the ground.
- Covers 2015-2020 only; it says nothing about later forest loss.
- It cannot tell the cause of loss (logging, fire, farming, natural dieback).

# Deliverables

## Files

| File | Type | Contents |
|---|---|---|
| rondonia_forest/forest_change.tif | GeoTIFF raster | Forest change |
| rondonia_forest/deliverables/forest_change_report.pdf | PDF report | Complete analysis report |
| rondonia_forest/deliverables/forest_change_report.xlsx | Excel workbook | Complete analysis report |
| rondonia_forest/deliverables/forest_change_report.docx | Word report | Complete analysis report |
| rondonia_forest/deliverables/maps/forest_change_map.png | PNG image | Map or chart for slides and documents |
| rondonia_forest/deliverables/maps/forest_change_chart.png | PNG image | Map or chart for slides and documents |
| rondonia_forest/deliverables/vectors/forest_change.gpkg | GeoPackage (all layers) | Vector features with attributes |
| rondonia_forest/deliverables/vectors/geojson/forest_change_aoi.geojson | GeoJSON, EPSG:4326 | Vector features with attributes |
| rondonia_forest/deliverables/vectors/geojson/forest_change_forest_change.geojson | GeoJSON, EPSG:4326 | Vector features with attributes |
| run_summary.json | JSON | Machine-readable record of this run |

## Vector layers

| Layer | Features | Total area (ha) | Note |
|---|---|---|---|
| forest_change | 36,685 | 23,134.3 | |

Vector layers made from rasters carry class_id, class_name, color_hex, area_ha, area_m2 (equal-area projection), pct_aoi, cx, cy, pipeline and period; layers a pipeline wrote itself (buildings, roads, streams, fire points) keep their own attributes. Coordinates are WGS 84 (EPSG:4326). The GeoPackage also holds the 'aoi' outline and, for raster-derived layers, a dissolved '<layer>_classes' summary.

# References


- Shimada, M. et al. (2014). New global forest/non-forest maps from ALOS PALSAR data (2007-2010). Remote Sensing of Environment 155, 13-31. (JAXA global PALSAR-2/PALSAR forest/non-forest map, Microsoft Planetary Computer.)